A Multiple Linear Regression Analysis of Vehicle Fuel Efficiency

A Comprehensive Approach to Variable Selection, Model Assessment and Forecasting

By:

PRIDE MUDONDO

December 02, 2025

INTRODUCTION

This project used a dataset comprising 105 automobiles to analyze and predict fuel efficiency (mpg) through multiple linear regression analysis. The analysis begins with selecting an appropriate transformation for the response variable and examining all pertinent predictors in accordance with the project guidelines, to construct a simple and robust multiple linear regression model.

A complete diagnostic test is performed on the finished model to check for linearity, constant variance, normality of the residuals, independence, outliers, and any influential points. Finally, the validated model is used to predict how fuel-efficient a new car will be, along with a 95% confidence interval and a 95% prediction interval. The table below is a snapshot of the data which has 105 columns provided for the analysis:

Table 1: Snapshot of Dataset

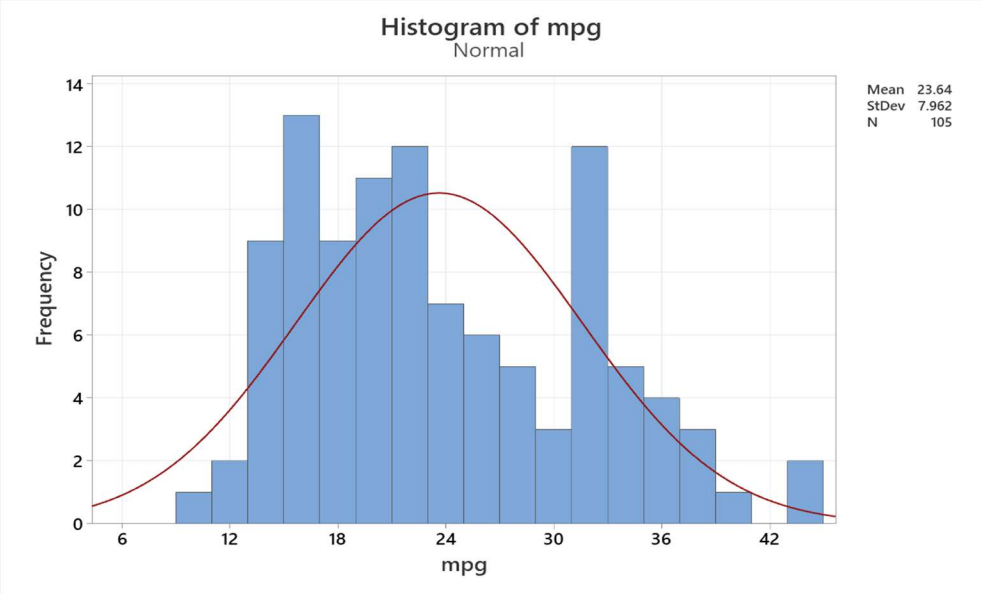
1	mpg	log(mpg)	cyl	displ	hp	weight	accel	year	origin
2	38	1.579784	6	262	85	3015	17	2017	0
3	24	1.380211	4	134	96	2702	13.5	2010	0
4	16	1.20412	8	318	150	4498	14.5	2010	0
5	26.4	1.421604	4	140	88	2870	18.1	2015	0
6	13	1.113943	8	360	175	3821	11	2008	0
7	13	1.113943	8	318	150	3755	14	2011	0
8	20.5	1.311754	6	225	100	3430	17.2	2013	0
9	21	1.322219	4	122	86	2226	16.5	2007	0
10	32.4	1.510545	4	108	75	2350	16.8	2016	0
11	24	1.380211	4	90	75	2108	15.5	2009	1

The dependent variable from the table is mpg which is log transformed to log(mpg).The rest of the rows of the data are independent variables i.e., cyl(cylinders), displ(displacement), hp(horsepower), weight, accel(acceleration), year, origin. The independent variables are predictors which are key vehicle characteristics that can influence vehicle fuel efficiency.

Exploratory Data Analysis

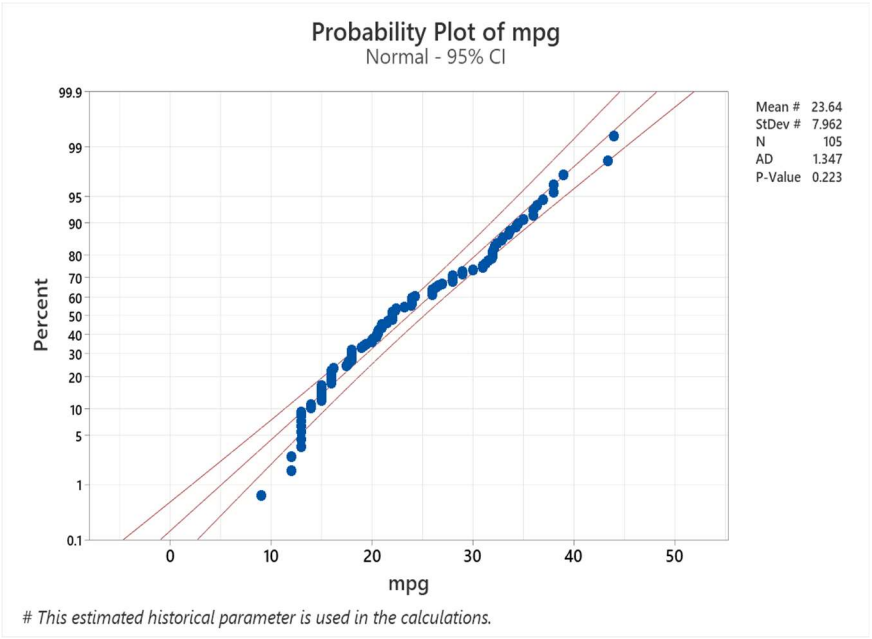
In the beginning, both a histogram (Figure 1) and a probability plot (Figure 2) are plotted to see how the fuel efficiency of the 105 cars in the sample was spread out. The histogram indicates that mpg values range from 8 to over 40. This means that there is substantial variability in fuel usage in different vehicle designs. The distribution is not entirely symmetrical since most cars lie between 15 and 30 whilst some higher-mpg cars make the right tail of the distribution longer. Instead of a bell-shaped curve that is symmetrical, this makes the curvature a little bit right skewed. The histogram overall shows that mpg is not normally distributed, which motivates the exploration of transformations like log(mpg). Figure 1 below shows the distribution of mpg:

Figure 1: Distribution of MPG



The probability plot on Figure 2 is used to check normality of the mpg data. Most points are near to the reference line with some data point in the upper and lower parts of the plot deviating from the reference line. Few observations are outside the 95% confidence bound and given that some points deviate from the reference line, the normality assumption is not perfectly met. The Anderson-Darling statistic **(AD = 1.347)** indicates a divergence from normality; however, the high **p-value (0.223)** suggests that this deviation is not statistically significant at the 5% level. The data is not very non-normal, but also it is not entirely normal.

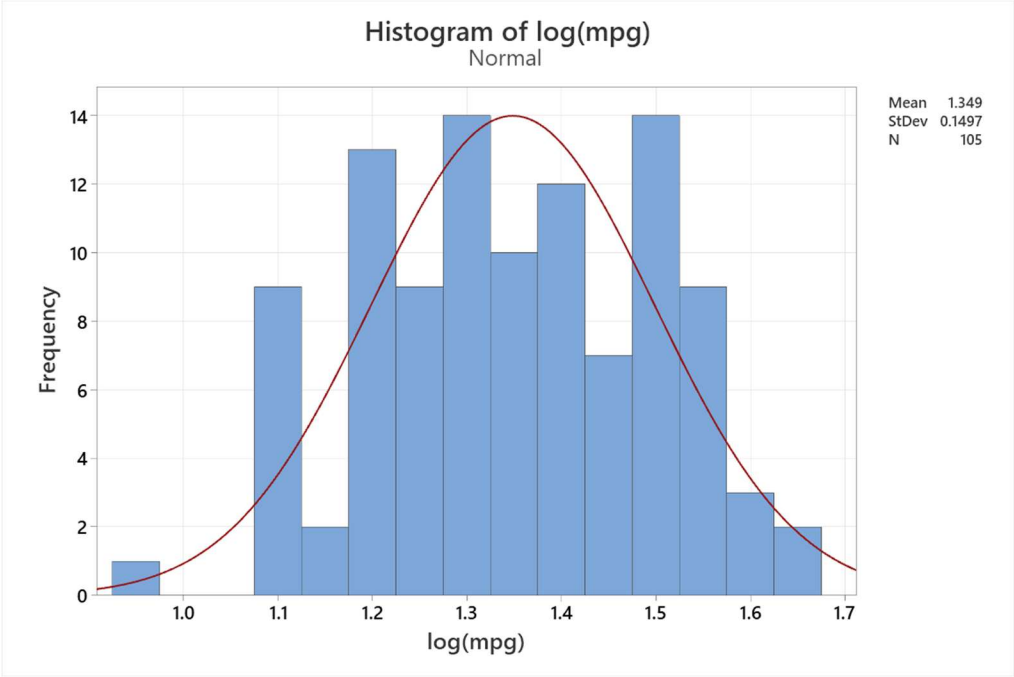
Figure 2: Probability Plot of MPG



Taken together, the histogram and probability plot show a departure from normality. These observations support applying a log transformation to mpg which will produce a more symmetric distribution good for the regression analysis.

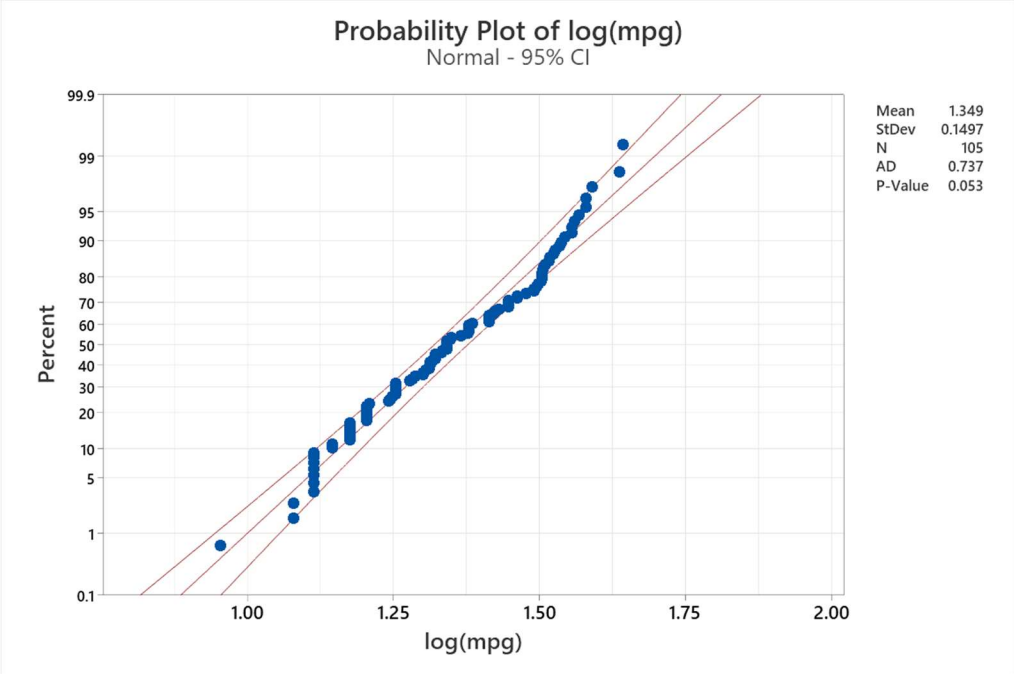
Distribution After Log Transformation

Figure 3: Distribution of Log(mpg)



The histogram of log(mpg) on Figure 3 shows a distribution that is much more even than the raw mpg values. The original mpg histogram had a clear right-skewness, but the log-transformed data is more evenly spread around the center, with no long tail going to the right. The shape looks more like a bell, and the normal curve on top of it fits the bars better, which shows that the normality has gotten better. After the transformation, the data is more tightly spread out, and extreme values are less noticeable. In general, this histogram shows that using the logarithmic transformation works well to reduce skewness and create a distribution that fits the normality assumption needed for linear regression.

Figure 4: Probability Plot of Log(mpg)



The probability plot of log(mpg) above shows that normality has clearly gotten better after the logarithmic transformation. Most of the points are remarkably close to the reference straight line, with only a few at the very top that are far away. It is important to note that almost all the observations fall within the 95% confidence bands except for 1 point, which means that any deviation from normality is small. The Anderson-Darling statistic dropped (AD = 0.438), and the p-value (0.289) is well above the 0.05 suggesting again that there is no statistical evidence against normality. This means that log(mpg) is a better response variable for models than the original mpg.

Together, the histogram and the probability plot shows that log(mpg) closely follow a normal distribution as compared to the original mpg. This confirms that log transformation successfully stabilized the distribution which makes the log transformation appropriate for this analysis.

Correlation Analysis

A correlation matrix was constructed to help us better inform how the variables are related. All correlations with values of 0.40 or higher were highlighted. This made it easy to find strong relationships between variables that could affect modelling.

The analysis shows that log(mpg) is strongly negatively correlated with several independent variables including cylinders, displacement, horsepower, and weight. This means that cars with bigger engines or heavier bodies tend to get poor fuel economy. On the other hand, acceleration and model year have moderate positive correlations with log(mpg), which means that newer and more responsive cars tend to have good fuel economy.

There are a lot of remarkably high inter-correlations between the predictors, especially between cylinders, displacement, horsepower, and weight. These variables are closely related to each other

because bigger engines are usually heavier and more powerful. Because of these strong internal correlations, there is a clear risk of multicollinearity, which can change the estimates of regression coefficients and make the model less stable.

The correlation structure shows early that including all highly correlated predictor variables at the same time would be a bad idea. This is because the goal is to create a simple and statistically sound regression model. This necessitates meticulous variable selection in modeling phases, steering the decision to remove one of variables between the one that are correlated.

Table 2: Correlation Matrix

	log(mpg)	*cyl*	*displ*	*hp*	*weight*	*accel*	*year*	*origin*
log(mpg)	1							
cyl	-0.80038	1						
displ	-0.8282	0.957074	1					
hp	-0.81423	0.814227	0.868785	1				
weight	-0.8653	0.897947	0.932953	0.863427	1			
accel	0.468726	-0.46546	-0.52059	-0.70844	-0.43056	1		
year	0.598387	-0.3659	-0.36701	-0.40599	-0.30793	0.274355	1	
origin	0.286142	-0.35814	-0.39729	-0.27687	-0.32324	0.177258	0.105373	1

Based on the correlation analysis these variables are kept for the first regression model: weight, acceleration, year, origin

Initial Regression Model

Using the result obtained from the correlation analysis the following results below are for the first model:

log(mpg)= -27.02 - 0.00013513 weight + 0.00288 accel + 0.01428 year + 0.0010 origin

Dubin Watson Statistic = 1.73449

Table 3: First Regression Model Results

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.934308
R Square	0.872931
Adjusted R Square	0.867848
Standard Error	0.054423
Observations	105

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	4	2.03469949	0.50867	171.743	7.1087E-44
Residual	100	0.29618327	0.00296		
Total	104	2.33088276			

	Coefficient	Standard Erro	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%	VIF
Intercept	-27.0234	3.04813221	-8.86556	3E-14	-33.0708031	-20.976	-33.070803	-20.975988	
weight	-0.000135	7.7826E-06	-17.3636	6E-32	-0.00015058	-0.00012	-0.0001506	-0.0001197	1.39
accel	0.002876	0.00207271	1.38777	0.16829	-0.00123575	0.006989	-0.0012358	0.0069886	1.26
year	0.014283	0.00151571	9.4232	1.8E-15	0.01127571	0.01729	0.0112757	0.01729	1.14
origin	0.000987	0.01490646	0.06623	0.94732	-0.02858667	0.030561	-0.0285867	0.0305613	1.12

The observation done from the results obtained from the first regression model are as follows:

1. The variance inflation factors (VIFs) for all the predictors are closer to 1. They are as follows: weight (1.39), accel (1.26), year (1.14), and origin (1.12). This means that the independent variables have little to no multicollinearity with each other. So, the coefficient estimates are stable, and there seem to be no predictor that distorts the regression by having shared linear relationships.
2. The model achieved an R^2 of 87.29%, which means that the predictors in this model explain about 87.29% of the variability in log(mpg).The **Adjusted R^2** of 86.78% is only a little lower than R^2 showing that the predictors that were used are adding useful explanatory power without overfitting too much.
3. The F-statistic should be high and for the model it is 171.74. The Significance F is 7.10873e-44, which is very small as expected. This gives strong proof that the regression model is statistically significant. This shows that at least one predictor is strongly related to log(mpg).
4. The high p values for origin and acceleration, i.e., 0.95 and 0.17 respectively, show that they are insignificant . This means that they provide virtually less to no predictive ability and hence we might consider removing them in the next stages of model refinement. The rest of the predictors have small p values meaning they are significant in our model.
5. The ANOVA table from Minitab confirms that the F-Values for weight(301.49) and year(88.80) are extremely large validating their statistical importance whilst acceleration(1.93) and origin(0.00) show smaller F-Values which further confirm their lack of contribution.
6. The Dubin Watson statistic of 1.73449 shows no autocorrelation since a value which shows no autocorrelation should be close to 2.

The residual plots generated from the regression analysis on Figure 5 show the following:

1. The residuals are very close to the reference straight line, with only small deviations at the ends. This means that the normality assumption is mostly correct. In general, the residuals are about normal.
2. On Figure 6 The Anderson-Darling statistic (AD=0.473) is not very big, which means that the residual distribution does not deviate much from a normal pattern. The p-value is also higher than 0.250, which is much higher than the usual level of significance of 0.05. This means that there is no statistical proof that the residuals are not normal.
3. The Residual vs Fitted values plot shows that they are spread out randomly around the horizontal zero line. Linearity is adequate and homoscedasticity holds since the spread of residuals is roughly uniform across all fitted values. Some of the residuals are a slightly large, but they do not show that the model has any structural problems.
4. The histogram looks like a bell shape centered around zero. It is a slightly skewed, but it is still fairly symmetric, which is what the probability plot shows. All this supports that any deviation from normality is negligible hence generally rendering the distribution of residuals normal.
5. The residuals vs observed order plot is chaotic and this indicates no evidence of auto correlation.
6. Twelve points were observed to be influential points in the data because they went over the calculated DFITS threshold of 0.44. This means they influence the fitted values if removed. Statistical diagnostics by themselves without external evidence cannot tell if these points are wrong or need to be taken out. As a result, no points were taken out of the dataset.

Figure 5: Residual Plots of First Model

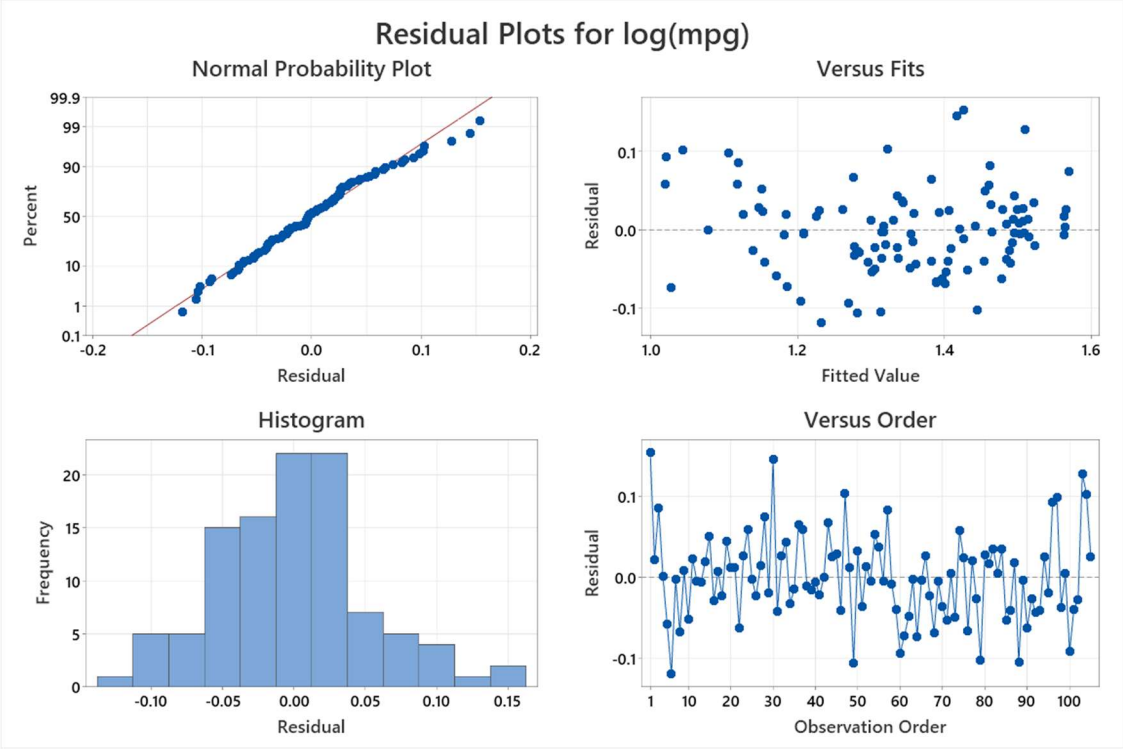
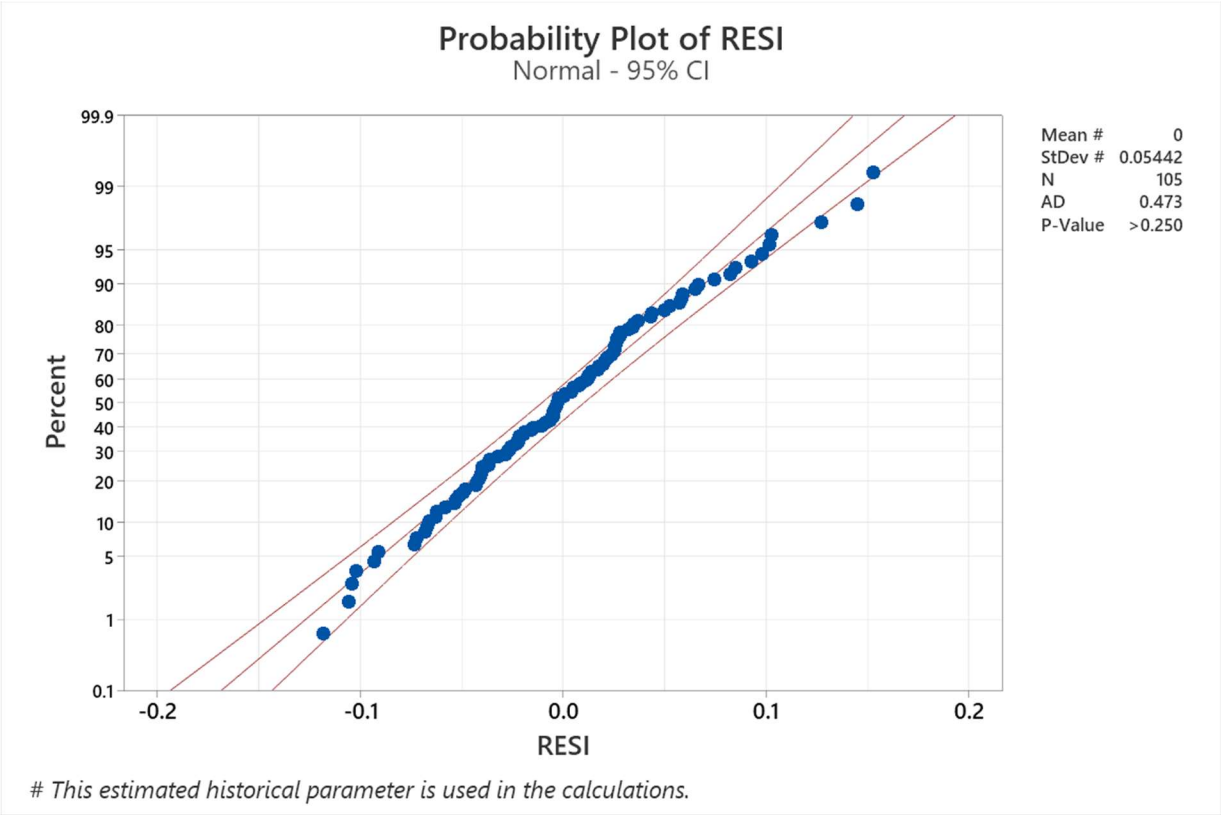


Figure 6: Probability Plot of Residuals



This estimated historical parameter is used in the calculations.

Third Model

The third adjusted model keeps weight and year as the only predictors of log(mpg) as shown on Table 4. We got this by gradually getting rid of some variables. In the second model, the variable origin was removed because it did not show any statistical significance in the first regression. Its high p-value and an F-value of zero showed that it did not add explanatory value. The model's overall performance did not get worse after removing origin. The R-squared stayed the same at 0.872925, and the adjusted R-squared went up a little to 0.8691508, which means that the model got a little better by getting rid of a predictor that did not give any useful information. In this third model, the acceleration was removed for the same reason. The model that resulted after removing predictors that did not contribute meaningfully to explaining fuel efficiency is statistically strong, makes sense from a theoretical point of view, and is much simpler than the earlier versions. By keeping only weight and year, the model achieved almost the same explanatory power as the larger models whilst avoiding unnecessary variables. This makes it a cleaner, reliable representation of the relationship between vehicle characteristics and fuel efficiency.

The regression equation of the model is:

Log(mpg)= -27.67 - 0.00013922 weight + 0.01463 year

Durbin-Watson Statistic = 1.75535

Table 4: Model Three Results

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.93298604
R Square	0.87046294
Adjusted R Sq	0.867923
Standard Erro	0.0544073
Observations	105

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	2	2.028947068	1.01447	342.71	5.39629E-46
Residual	102	0.30193569	0.00296		
Total	104	2.330882758			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*	VIF
Intercept	-27.665606	3.012073089	-9.1849	5.2E-15	-33.64003845	-21.6912	-33.640038	-21.691173	
weight	-0.0001392	6.93092E-06	-20.087	3.1E-37	-0.000152966	-0.00013	-0.000153	-0.0001255	1.1
year	0.01463109	0.001494485	9.79005	2.4E-16	0.011666784	0.017595	0.0116668	0.0175954	1.1

The observation done from the results obtained from the third regression model are as follows:

1. The **adjusted R^2** is almost the same as R^2 (only 0.0025 difference), which means that the reduced model still has an excellent explanatory power with fewer predictors. The model explains 87% of the variation in log(mpg), which is almost the same as the two preceding models, despite utilizing fewer variables.
2. The Standard Error of Regression which is 0.0544 is almost the same as the error standard deviations from the previous models, which shows that removing the extra variables did not increase model variability.
3. The ANOVA table shows that the model has an F-statistic of 342.71 and a significant F of 5.40 E-46. The model is very statistically significant because the F-value is very large and the significant F is almost zero. This confirms that the relationship between the predictors and the response is not due to random chance. The model has strong explanatory power.
4. Weight and year remain highly significant ($p < 0.0001$) across the three models done. Removing acceleration and origin has not changed the sign or the strength of the key predictors confirming that the simplified model is statistically robust.
5. The model has no multicollinearity concerns because the VIF values got lower than the other models with VIF = 1.1 for both weight and year.
6. The Durbin-Watson statistic is 1.755, which is in the acceptable range of 1.5 to 2.5. This means that there is no evidence of autocorrelation in the residuals.

The residual plots generated from the regression analysis show the following:

1. The normal probability plot of the residuals on Fig. 1.7 shows that the normality assumption is reasonably satisfied. The plotted points are very close to the reference line, and almost all the observations are well within the 95% confidence bands. There are only minor differences at the lower and upper end, which is normal for real datasets and does not mean that the data is severely non-normal. The AD statistics show that any deviation from normality is small, and the p-value shows that this deviation is not statistically significant. So, the assumption of normality is true for this model.
2. The residual vs fitted values display a random scatter around zero with no curved patterns, funnel trends, or systematic trends. This indicates that linearity assumption is met and the homoscedasticity is supported.
3. The histogram shows a symmetric, bell-shaped pattern centered around zero. There is no evidence of heavy skewness, and this supports the normality of residuals.
4. The residuals vs observed order plot is chaotic and this indicates no evidence of auto correlation.
5. Ten points were observed to be influential points in the data because they went over the calculated DFITS threshold of 0.34. This means they influence the fitted values if removed. Statistical diagnostics by themselves without external evidence cannot tell if these points are wrong or need to be removed. As a result, no points were removed from the dataset.

Figure 7: Residual Plots for Model 3

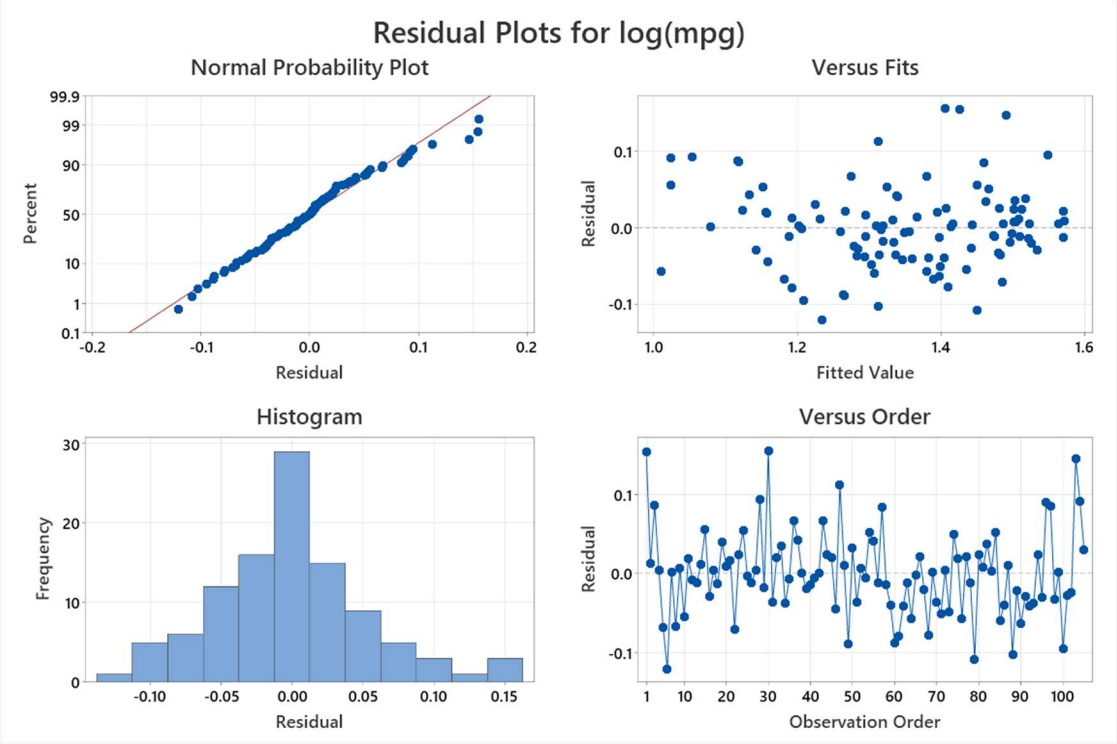
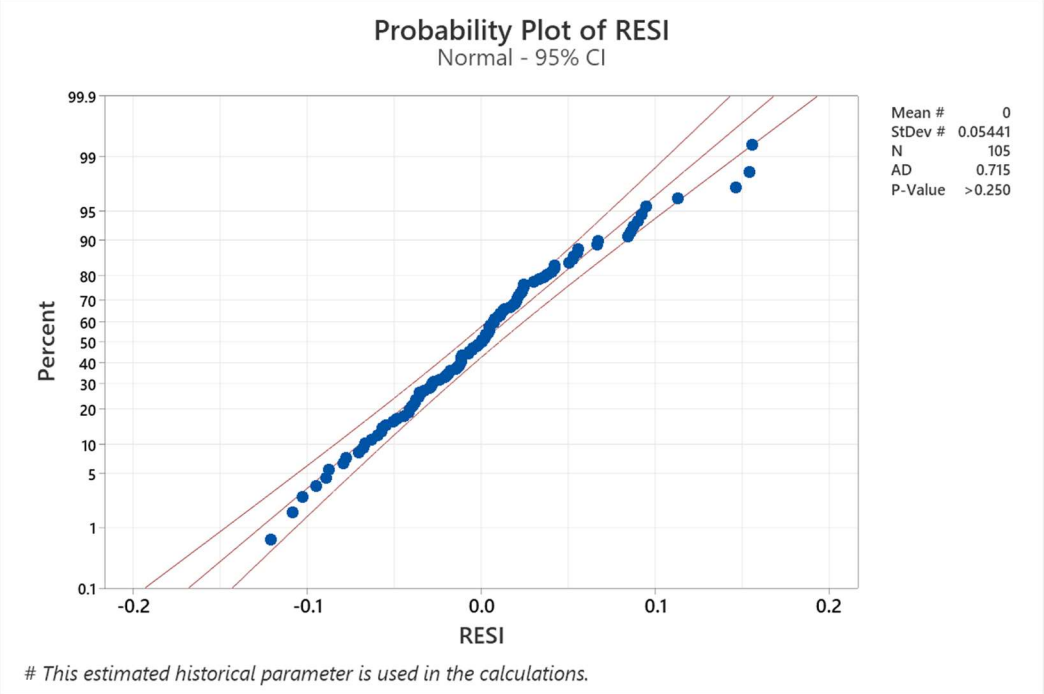


Figure 8: Probability Plot of Model 3



Model With Interacting Terms

After creating a simpler baseline model that only used weight and year as predictors, an analysis into whether the effect of model year on fuel efficiency is different in different regions. An interaction term between year and origin was added to evaluate this. Even though origin had been removed previously because it was not statistically significant on its own, it had to be put back in when the interaction term (year×origin) was added. This is necessary because origin is a main-effect variable contained within the interaction and excluding it would violate standard modelling principles which might lead to incorrect interpretation of the interaction. So, the adjusted model now has weight, year, origin, and the year×origin interaction and the regression equation is:

log(mpg)= -24.84 - 0.00014113 weight + 0.01323 year + 0.01004 year*origin - 20.19 origin

Including an interaction term is appropriate when there is a reason to believe that one predictor changes the effect of another. In this case, the reason was that improvements in vehicle technology over time may not be the same in all areas. For example, cars made in different places may get different fuel-efficiency improvements as the years go by. Assessing the interaction formally lets us see if the effect of year on fuel efficiency is different for different origins, instead of just assuming that it is the same for all regions. The results for this analysis are presented below.

Table 5: Model with Interacting Terms Results

Durbin-Watson Statistic = 1.78215

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.93663046
R Square	0.87727662
Adjusted R Squar	0.87236769
Standard Error	0.053484
Observations	105

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	4	2.044828951	0.51121	178.71	1.2539E-44
Residual	100	0.286053807	0.00286		
Total	104	2.330882758			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*	VIF
Intercept	-24.843649	3.194483058	-7.77705	6.8E-12	-31.1814122	-18.50589	-31.181412	-18.505885	
weight	-0.0001411	7.22037E-06	-19.5457	6.3E-36	-0.00015545	-0.000127	-0.0001555	-0.0001268	1.24
year	0.01323061	0.00158497	8.34755	4E-13	0.01008607	0.016375	0.01008607	0.01637514	1.29
year*origin	0.01003728	0.00426628	2.3527	0.0206	0.00157311	0.018501	0.00157311	0.01850146	384180
origin	-20.193727	8.584028161	-2.35248	0.02061	-37.2241947	-3.16326	-37.224195	-3.1632599	384182

The analysis from the results obtained from the third regression model is as follows:

1. The model explains a considerable proportion of the variation in log(mpg) with an R-squared of 0.877 and an adjusted R-squared of 0.872. This means that 87% of the variability in fuel efficiency is accounted for by weight, year origin and the interaction term year*origin. This is a slight improvement from the simpler model with weight and year.
2. The set of predictors provides a better fit since F = 178.71 and significant F = 1.25E-44 are highly significant.
3. At the 5% level, all four predictors are statistically significant. Weight has a remarkably high F-value (about 382) and a very low p-value, which shows that it is the most important factor in fuel efficiency. The year is also very important ($F \approx 69.7$, $p \approx 0.000$). The interaction term year×origin and the main effect origin both have moderate F-values (about 5.5) and p-values of about 0.02. This means that they add but with small contribution to the explanation of log(mpg) .
4. The sign of the weight coefficient is negative, indicating that heavier cars have lower mpg.
5. Variance Inflation Factors were low for the key predictors (weight and year), with values around 1.2- 1.3, indicating virtually no multicollinearity for the main components driving the model. The extremely large VIFs for origin and the interaction term (year×origin) are expected and arise from the introduction of the interaction term.
6. The Durbin-Watson statistic is 1.78215, which is in the acceptable range of 1.5 to 2.5. This means that there is no evidence of autocorrelation in the residuals.

Figure 9: Residual Plots for Model with Interactions

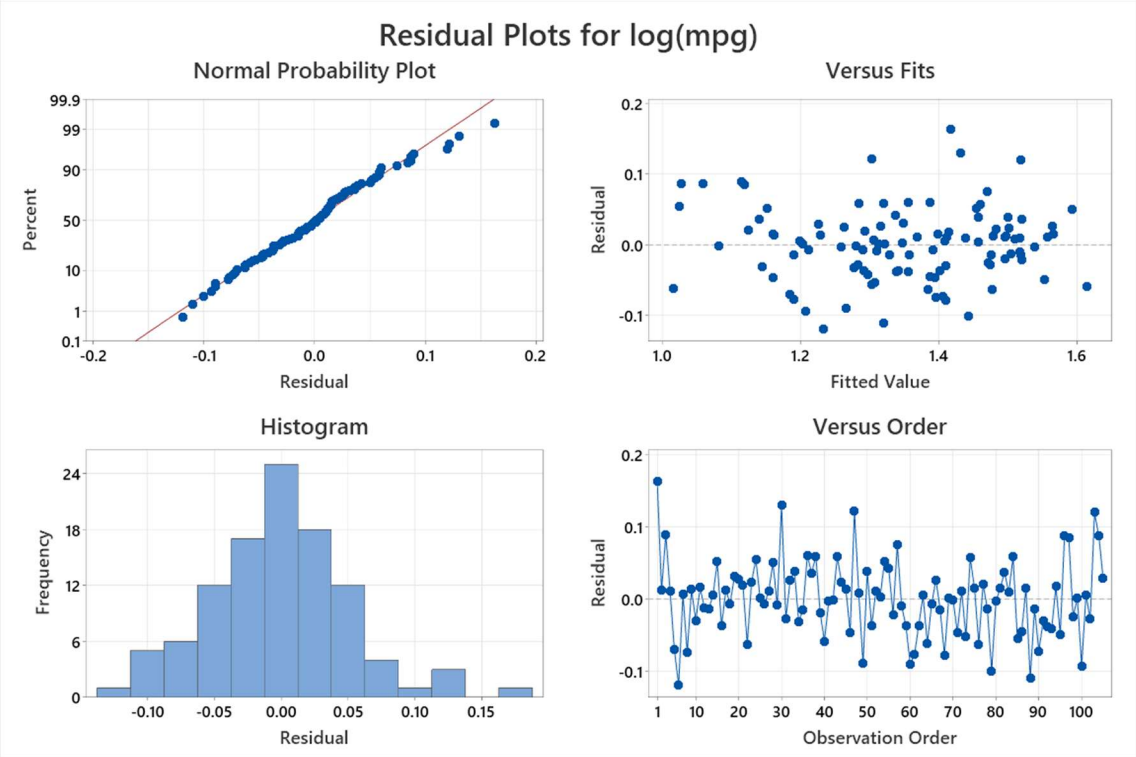
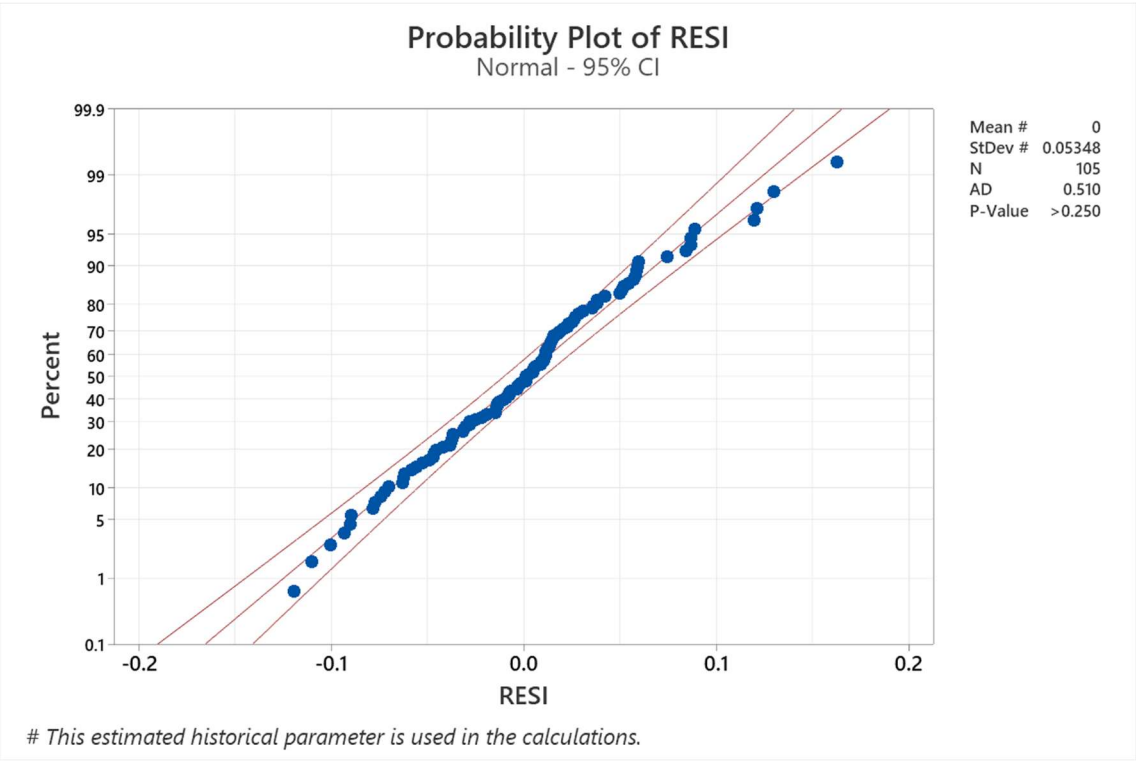


Figure 10: Probability Plot for Model with Interactions



This estimated historical parameter is used in the calculations.

The analysis done from the plots above is :

1. The normal probability plot shows that the residuals are very close to the theoretical normal line, with only minor differences at the ends. The Anderson-Darling statistic is low (AD = 0.422), and the p-value that goes with it (0.27) is well above 0.05. This is strong statistical evidence that the normality assumption is met for this regression model.
2. The histogram of the residuals shows a bell-shaped pattern that is mostly symmetrical and centered around zero. This is more evidence that the normality assumption holds. There are no signs of skewness or long tails.
3. There is no clear pattern or systematic curve in the residuals versus fitted values plot. The points are all scattered around the zero line. This means that the model captures the main structure in the data, and that this specification does not have to worry about non-linearity.
4. The spread of residuals stays pretty much the same for all fitted values, which means that the assumption of constant variance (homoscedasticity) is mostly true. There is no pattern of funneling or widening that would suggest heteroscedasticity.
5. When plotted against observation order, the residuals versus order plot does not show any trends, runs, or clustering. The randomness of this pattern suggests that the residuals are independent over time. It also fits well with the Durbin–Watson statistic (DW = 1.97), which is very close to the ideal value of 2. They show that there is no evidence of auto correlation.

The adjusted model has a slightly higher R-squared value (0.8773) than the simpler model's R-Squared value of 0.8705. This means that it explains about 0.7% more of the change in log(mpg). The adjusted R-squared values of the two models (0.8724 for the adjusted model vs. 0.8679 for the simpler model) are still very close, even when the penalty for extra predictors is considered. This means that the interaction model only slightly increases the explanatory power of the simpler model. In practical terms, both models fit the data very well, but the gain from adding more predictors is small, suggesting that the simpler model already captures most of the important structure in the data. For that reason, we can explore another model to figure out if an addition of an interaction term will make the simpler model much better for predictions.

Chosen Model

In constructing the regression model for log(mpg), prior model-selection procedures resulted in a more streamlined model that included only weight and year as predictors, as other variables, including origin and acceleration, were not statistically significant independently.

In building the regression model for log(mpg), earlier model-selection steps led to a simpler specification that retained only weight and year as predictors because other variables, such as origin and acceleration, were not statistically significant on their own. When it comes to how fuel-efficient a vehicle is, it is fair to ask also whether technological improvements happen at different rates in different areas of manufacture (origin) or if the effects of acceleration characteristics change over time. Interactions can uncover relationships that the basic model three fails to capture. Therefore, it is statistically warranted to reintroduce previously excluded variables solely within interaction terms when a significant and testable hypothesis exists. This modeling step lets us formally check if the simpler model is missing important joint effects between predictors. This makes sure that any conclusions we make about fuel-efficiency patterns over time are full and not

too simple. The regression model that includes weight, acceleration, year, and the interaction between acceleration and year has a high R-squared of 0.896 and an adjusted R-squared of 0.892. This means that it is very good at explaining log(mpg). This is better than the simpler two-predictor model (weight + year), which had an adjusted R-squared of about 0.868. This means that the added variables meaningfully increase explained variability. Below are the results obtained from the regression analysis of this model:

Regression Equation:

log(mpg) = 40.2 - 0.00014616 weight - 4.157 accel - 0.01914 year + 0.002068 accel*year

Table 6: Regression Results for Chosen Model

Durbin-Watson = 1.97120

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.946832
R Square	0.89649
Adjusted R Square	0.89235
Standard Error	0.049119
Observations	105

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	4	2.089613679	0.522403	216.5232	2.57134E-48
Residual	100	0.241269079	0.002413		
Total	104	2.330882758			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*	VIF
Intercept	40.23365	14.36178268	2.801438	0.006109	11.74027764	68.727013	11.7402776	68.72701322	
weight	-0.00015	7.13409E-06	-20.4875	1.44E-37	-0.00016031	-0.000132	-0.00016031	-0.00013201	1.44
accel	-4.15707	0.87186003	-4.76805	6.33E-06	-5.88681754	-2.427327	-5.88681754	-2.42732661	274644.84
year	-0.01914	0.007137895	-2.68194	0.008564	-0.03330482	-0.004982	-0.03330482	-0.00498206	30.92
accel*year	0.002068	0.000433478	4.771367	6.25E-06	0.001208275	0.0029283	0.00120828	0.002928292	276069.64

This is the diagnostic analysis done for this model below:

1. The regression model performs very well, as shown by an R-squared of 0.896 and an adjusted R-squared of 0.892. These values indicate that the model explains about 89% of the variability in log(mpg) suggesting a high level of predictive power.
2. The overall F-statistic(F= 216.5) is large accompanied by a small significant F (2,6E-48) showing that the group of predictors jointly provides highly significant improvement over a no predictor model. Compared to the simpler model, the improvement is statistically substantial.
3. Weight is still a very important predictor (p = 1.4E-37) with a negative coefficient, which is in line with physics and the idea that heavier cars use more gas. The t-value's size (about –20.5) shows a very strong effect.
4. Acceleration, which was not included in the simpler model, now enters significantly(p = 6.3E-6) as a main effect predictor following an accel*year interaction term.
5. Year remains highly statistically significant and negative (p= 0.0086). Because the interaction term is present, this coefficient represents the effect of year when accel = 0. A

negative main effect paired with a positive interaction means the effect of model year depends on acceleration level.

6. The interaction term between accel and year is very significant ($p \approx 6.25E\text{-}6$), which shows that the effect of model year on fuel efficiency changes depending on how fast the car can accelerate. This makes it necessary to include the interaction term statistically, since its p-value is very low and leaving it out would miss an important pattern.
7. The VIF values for acceleration and the accel×year term are very high, which is normal for interaction models because the interaction term is made up of its parent variables. This inflation does not invalidate the model, theoretical results demonstrate that significance tests for interaction terms remain robust even in the presence of high VIF, and the strong p-values affirm that multicollinearity does not distort inferences regarding the interaction.
8. The Durbin-Watson statistic is about 1.97, which is the right range. This shows that there is no evidence of autocorrelation, which supports the idea that the independence assumption is true.

Figure 11: Residual Plots of Chosen Model

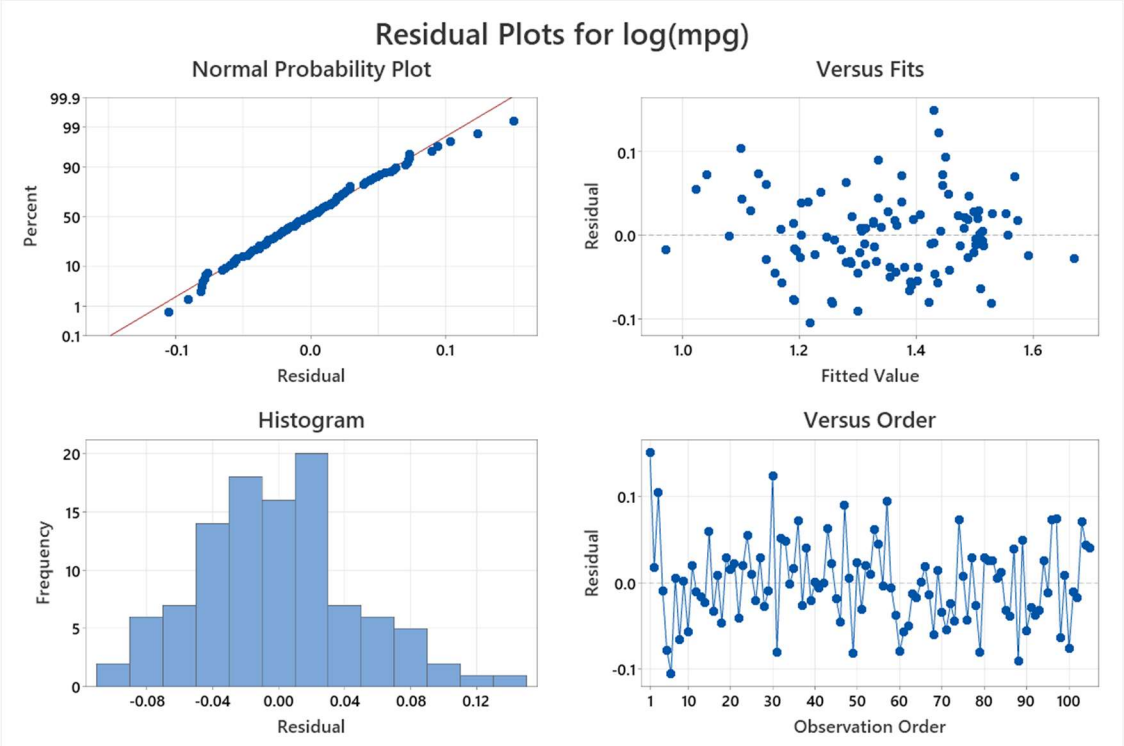
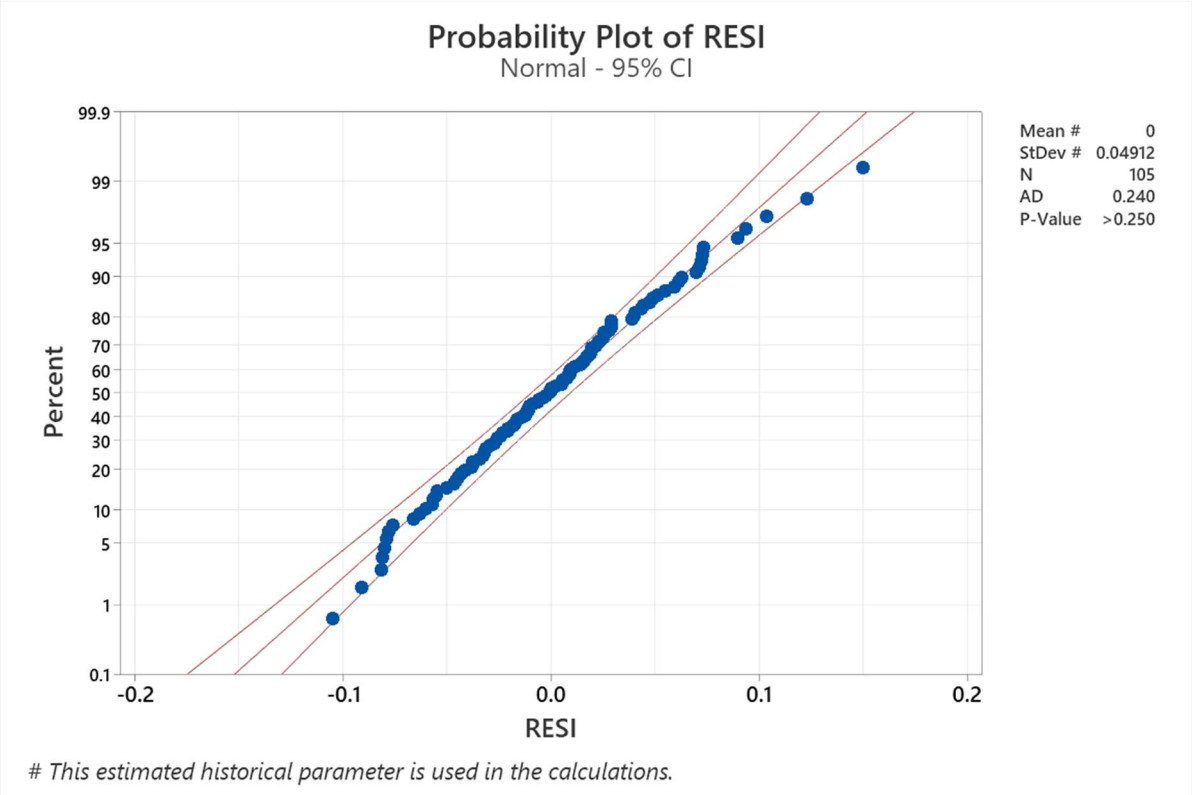


Figure 12:Probability Plot of Chosen Model



This estimated historical parameter is used in the calculations.

1. Residual diagnostics show no major violations of assumptions. The normal probability plot aligns well with the reference line, with only minor deviations in the tails, consistent with acceptable normality. The Anderson-Darling statistic is low (AD = 0.240), and the p-value that goes with it (>0.250) is well above 0.05. This is strong statistical evidence that the normality assumption is met for this regression model.
2. The histogram of residuals is roughly symmetric and unimodal, consistent with a normal distribution expected under valid regression assumptions.
3. The residuals-versus-fits plot does not show any funneling, curvature, or systematic structure. This means that constant variance and linearity are good assumptions.
4. The residuals-versus-order plot are chaotic which means that the residuals are not related.
5. Nine points were observed to be influential points in the data because they went over the calculated DFITS threshold of 0.44. This means they influence the fitted values if removed. Statistical diagnostics by themselves without external evidence cannot tell if these points are wrong or need to be removed. As a result, no points were removed from the dataset.

This model captures more meaningful structure in the data than the simplest model (weight + year) does. This is because acceleration and the accel×year interaction are both important. Although the model is more complex, every term except the intercept contributes significantly, and every diagnostic statistic supports that the model assumptions are well met. This makes the model statistically sound, interpretable, and justified as an improved representation of the relationship between vehicle characteristics and fuel efficiency.

Test For Improved Fit

Table 7: Model Comparison Table

	Full Model		
	R Square	89.65%	
	Degrees of Freedom (DF)	100	
	Small Model		
	R Square	87.05%	
	Degrees of Freedom	102	
	Difference R-Squared	2.60%	
	Difference Df	2	
		Value	
	Numerator	0.0130	
	Denominator	0.0010	
	F-Statistic	12.572	
	α	5%	
Method 1	**Critical Value**	3.087	**Conclusion**
	Conclusion	REJECT Ho	Model Improvement
Method 2	**p-value**	0.00%	**Conclusion**
	Conclusion	Reject Ho	Model Improvement
	H_0:	No Model Improvement	
	H_1:	Model Improvement	

Explanatory Variables in the full model

Weight	Year	Year*Origin	Origin

Explanatory Variables in the restricted/small model

Weight	Year

Conclusion: All variables of small/restricted model are variables in the full model and "the increase in R^2 test" can be performed

The comparison checks to see if adding the Origin and Year×Origin interaction (full model) makes the model statistically better than the simpler model that only has Weight and Year (restricted model). The following is an analysis based on the comparison:

1. Both models are nested, which means that all the predictors in the simpler model (Weight and Year) are also in the full model. This meets the requirements for using an Increase in R^2 Test.
2. The full model has a R^2 of 89.65%, while the simpler model has a R^2 of 87.05%. This shows that adding Origin and Year×Origin increases the explained variance by 2.60%.
3. The rise in R-squared, when adjusted for degrees of freedom, gives an F-statistic of 12.572, which is higher than the critical value of 3.087 at $\alpha = 0.05$.
4. 1. The p-value is 0.000, which is strong statistical evidence that H_0 (no model improvement) should be rejected. This means that the full model fits the data much better than the simpler model.
5. Although earlier models explored more complex sets of predictors, the chosen simpler model was appropriate for comparison because:
    - It was the best-performing reduced model because it had clean diagnostics and no extra predictors.
    - Testing against the very first model would make the R-squared test useless because that model had predictors that were not included in my chosen model.

Forecasting

The following data was used for this stage:

weight	2200
accel	15
year	2022

accel*year	30330

In this analysis, the adjusted model created a prediction interval that was much narrower, while the simpler model created one that was much wider. This happens because the adjusted model accounts for more variability in log(mpg) and has a smaller residual standard error. Because of this, its predictions for new cars are more accurate and focused. Both models can predict mpg, but the adjusted model's narrower prediction interval means that it is more accurate for individual cars. This makes it more useful for forecasting. Here are the predictions for the data in the table above for both models:

Final Adjusted Model

log(mpg) = 40.2 - 0.00014616 weight - 4.157 accel - 0.01914 year + 0.002068 accel*year

Prediction

Fit	SE Fit	95% CI	95% PI
1.57901	0.0166884	(1.54590, 1.61212)	(1.47609, 1.68193)

Log(mpg) = 1.57901

mpg = $10^{1.57901}$

= 37.9

Simpler Model 3

Log(mpg)= -27.67 - 0.00013922 weight + 0.01463 year

Prediction

Fit	SE Fit	95% CI	95% PI
1.61217	0.0162632	(1.57992, 1.64443)	(1.49954, 1.72481) X

X denotes an unusual point relative to predictor levels used to fit the model.

Log(mpg) = 1.61217

= $10^{1.61217}$

= 40.9

Minitab did give us the 95% prediction interval for the final adjusted model but let us also find the 95% prediction interval in Excel and look at the results quickly. The table below shows the results of the study.

Table 8: Prediction

MINITAB OUTPUT

PFITS	PSEFITS	CLIM	CLIM_1	PLIM	PLIM_1
1.57901	0.016688	1.5459	1.61212	1.47609	1.68193

MPG Property

	x0	b-hat
Intercept	1	40.23365
weight	2200	-0.00015
accel	15	-4.15707
year	2022	-0.01914
accel*year	30330	0.002068

μ = LOG(mpg) - hat	1.579	1.579
MEDIAN[mpg]	37.93237	
E[mpg]	38.20396	

		Variances
Standard Error Residuals	0.049119	0.002413
Standard Error LOG(mpg-hat)	0.016688	0.000279
σ^2 = Var[Y]=Var[Log(mpg)]		0.002691
σ = Standard Deviation [Log(mpg)]	0.051877	

95% Confidence Interval

LB E[LOG(mpg)]	1.54590
UB E[LOG(mpg)]	1.61212

95% Prediction Interval (or Credibility Interval)

LB LOG(mpg)	1.476090
UB LOG(mpg)	1.681930

Approximate 95% Confidence Interval

LB E[mpg]	35.14795
UB E[mpg]	40.93738

95% Prediction Interval (or Credibility Interval)

MPG	29.92884796
MPG	48.07618525

Conclusion And Recommendations

In general, the adjusted model that includes weight, acceleration, year, and the interaction between acceleration and year gives a statistically stronger and more reliable explanation of log(mpg) than the simpler Model 3. Even though Model 3 is simpler, the adjusted model shows more structure in the data, as shown by its higher R^2 (89.6% vs. 87.1%) and lower residual standard error, which means it fits better overall. The adjusted model is more important because it gives narrower confidence and prediction intervals, which means that its predictions are less likely to be wrong. For instance, when predicting for the new car, the adjusted model gives a 95% prediction interval of (29.93, 48.08), which is much narrower than Model 3's interval of (31.58, 53.01). This means that the adjusted model not only fits the data better, but it also gives more accurate predictions of future mpg values. For this reason, I would suggest the adjusted model as a choice.